

Quenton Ni

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EDUCATION

University of Minnesota, College of Science and Engineering

M.S. in Computer Science | GPA: 4.00

B.S. in Computer Science, Minor in Management Information Systems | GPA: 4.00

Minneapolis, MN

Expected May 2027

May 2026

EXPERIENCE

IT Intern

Land O' Lakes

Arden Hills, MN

May 2026 – July 2026

- Architected a web scraping agent using Azure AI Foundry to extract and normalize K-12 meal data across 100+ districts, eliminating an estimated 120 hours of manual sales research while equipping the sales team with new market intelligence.
- Fine-tuned and evaluated RF-DETR and YOLO object detection models to identify pallet defects and surface potentially disputable claims for manual review, delivering working prototypes for a claims process associated with \$3M in annual losses.

Software Engineer (Capstone)

Medtronic

Minneapolis, MN

Sep 2025 – May 2026

- Developed backend functionality for a FastAPI and React application analyzing logs from cardiac device management software, linking flagged logs to known Azure DevOps bugs through checksum based matching.
- Optimized bug-matching workflows using concurrent work-item retrieval, WIQL queries, and cached checksums; enabled engineers to confirm, revise, or dismiss suggested matches.
- Integrated automated unit tests and SonarQube analysis into Azure DevOps pipelines; collaborated across four student teams and a research lab to deliver a working prototype.

Research Assistant | NSF REU & UROP Grant Recipient

University of Minnesota-Twin Cities

Minneapolis, MN

Sep 2025 – May 2026

- Redesignated a ventricular arrhythmia classification model using depthwise separable convolutions, reducing its parameter count from 1,534 to 390.
- Implemented quantization-aware training with INT4 precision for ventricular arrhythmia classification, contributing to a 31X reduction in theoretical model footprint while retaining an F-beta score of 0.956 versus 0.964 for the FP32 baseline.

Computer Science / Machine Learning Teaching Assistant

University of Minnesota-Twin Cities | Data Science and AI Hub

Minneapolis, MN

Sep 2024 – May 2026

- Led weekly labs and office hours covering Java, C systems programming, and x86-64 assembly, supporting 400+ students and testing course materials.
- Supported PhD-led instruction for 90+ high school seniors, teaching machine learning concepts including deep neural networks, transformers, and reinforcement learning.

PROJECTS

Multi-Dialect Vietnamese Text to Speech

- Fine-tuned and evaluated a multi-dialect Vietnamese TTS system using F5-TTS on ~71 hours of dialect-annotated speech across Northern, Central, and Southern Vietnamese.
- Built and presented an interactive Gradio demo for dialect controlled Vietnamese speech synthesis with generated audio playback and mel spectrogram visualization.

Gamblr

- Engineered a real-time tracking application using Vue.js and Firebase Firestore, featuring a customizable drag and drop dashboard for live spending analysis and habit tracking.
- Enhanced user interaction by integrating Google Cloud Speech-to-Text via Firebase Functions and rendering an interactive 3D slot machine simulation using Three.js.

Distributed Auction Marketplace System

- Built a container based distributed auction marketplace using Docker Compose, with a centralized controller, stateless service nodes, and replicated SQLite backed storage nodes.
- Implemented primary-backup replication with controller managed health checks and automatic primary failover to maintain availability during storage node failures.
- Designed autoscaling for the service node layer using rolling request rate monitoring and round robin routing.

TECHNICAL SKILLS

Languages: Python, C, Java, JavaScript, TypeScript, HTML, CSS

Technologies: PyTorch, NumPy, scikit-learn, Transformers, Hugging Face, Git/GitHub, Docker, Vue, React, Flask, FastAPI, Azure, Azure DevOps, SQL, PostgreSQL, REST APIs, Agile/Scrum

Relevant Coursework: Applied Machine Learning, Natural Language Processing, Algorithms and Data Structures, Operating Systems, Machine Architecture, Distributed Systems